

Intuition behind complex numbers

$\mathbb{P} := \mathbb{R}_{\geq 0}$ has one “piece” of information: magnitude. if $f : X \rightarrow \mathbb{P}$, every point is assigned a magnitude. If we take a real function $f : X \rightarrow \mathbb{R}$, then each point is assigned a magnitude and a “polarity”. The rule of the polarity is that it is canceled out by its twin, i.e. the familiar rule:

$$a + (-a) = 0$$

The complex numbers push this analogy to its algebraic limit.

As we are thinking of $\mathbb{C}$ as one-dimensional, we must treat its points as being one point with two properties.

Let $a, b \in \mathbb{C}$. Then they are both of the form $x + iy = re^{i\theta}$. Let us think of each representation and what it means. Starting with the “natural” one: $z = re^{i\theta}$. It is “natural” in the sense that $e^{i\theta}$ are the natural units in in so many fields. This means that a single point z have magnitude information as well as *phase* information, where when we say phase we mean it like in the sentence *what phase are we in right now?* So $z = i$ is a quarter way through it’s phase. Any cyclic group $C_n \subseteq U(1)$ represents a set of discrete phases. Phases always must be able to return to the origin even when moving forward (i.e. they are periodic).

If we take $z = x + iy$, then then this is a little more. First, r is an “absolute force”. This force applies in two ways: direction and rotation (or phase). If x is large, the weight is height is high. if y large, then the “push towards phase change” is high. Then θ represents the ratio between these two forces. So far, I’m interpreting it like this:

1. x represents a change in the forces of a system, while y represents how much force is stable. For example, an orbiting satellite in a stable orbit can be thought of as having $x = 0$ but $y > 0$ so we have $y/0$ (i.e. Undefined) which makes sense as there is no ratio to be had. But $|x^2 + y^2|^{1/2} = |y|$ which shows there *is* energy in the system

So now, if we add two complex numbers, they are “independent” (or orthogonal): $a_1 + a_2y + b_1 + b_2y = (a_1 + b_1) + (a_2 + b_2)i$. Here is the thing. These “states” also represent a natural action. After coupling all the actions together and getting a number $x + iy$, we can then think of this number as being able to act on a any other “state”. For example, i acts on 2 by rotating or, or equivalently, 2 acts on i by scaling it. But $i + 2$ is just adding together actions without performing them yet. What distributivity tells us is that:

$$z(u + w) = zu + zw$$

So we can couple sets and then act on them, or act on them and then couple the results, in either case the result is the same.

Complex numbers are special in having the $zw = wz$ and associative and distributive “symmetry” that work well with coupling that is also associative and commutative. The “why” can probably be answers by doing some matrix-algebra and showing uniqueness.